

Supplementary Material

Network-based prioritization of venom–host interactions potentially relevant to Russell’s viper–associated thrombotic microangiopathy

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Contents: Supplementary Methods (SM.1–SM.9) · Tables S1–S10 (captions below; full machine-readable tables in the companion workbook Supplementary_Tables_S1-S10.xlsx) · companion data files note · Figure 5 vector export (Figure5_4layer_network.svg, included in this package).

Compiled 2026-10-02 (analysis change log v47). The analysis code and result tables are archived at <https://doi.org/10.5281/zenodo.23065792> (concept DOI, resolves to the latest release); large raw files at <https://doi.org/10.5281/zenodo.23087108>.

Supplementary Methods

SM.1 Venom-toxin library construction (Phase 1A)

Retrieval (all 2026-09-22, UniProtKB release 2026_03): main query keyword:"Toxin" AND reviewed:true AND organism_name:"Daboia russelii" returned 52 entries (28 *D. siamensis* covered by automatic synonym expansion; an independent *D. siamensis* export of 28 entries overlapped completely). Because the "Toxin" keyword misses the Kunitz inhibitor family, a supplementary query (organism_name:"Daboia russelii") AND (protein_name:"Kunitz") AND (reviewed:true) recovered 24 reviewed Kunitz entries, plus 4 CRISP entries (full-length CRISPs unreviewed, flagged in the source column). SVMP coverage was replenished from NCBI Protein with russelysin (GenBank AAZ39880) and two disintegrins. Merging and de-duplication by accession gave 78 records; 16 fragmented sequences <30 aa were removed by design (mostly short Kunitz/CRISP peptides), yielding 62 entries; the VenomZone cross-check recovered one full-length basic PLA2-3 (P86368), giving the final 63-entry library.

Quality control against VenomZone/Tox-Prot (query taxonomy_id:{8707/343250} AND cc_tissue_specificity:venom AND reviewed:true; 71 reviewed venom entries): the 21 entries absent from our library were 20 short-peptide sequencing-evidence entries (<30 aa, excluded by design) plus P86368 (recovered); seven reviewed entries present in our library but lacking a venom-tissue annotation were retained as reasonable additions. All five major families (SVMP/SVSP/PLA2/snaclec/Kunitz) are represented in both resources.

Redundancy was reduced by 90%-identity clustering (custom greedy CD-HIT-equivalent; script build_toxin_lib.py), giving 40 clusters with 40 representative sequences (toxins_nr90.fasta). Family-level venom abundance weights were back-filled from the functional venomomics dataset of Sri Lankan *D. russelii* (Tan et al., J Proteomics 2015): PLA2 35.0%, snaclec 22.4%, SVSP 16.0%, SVMP 6.9%, LAAO 5.2%, Kunitz 4.6%, NGF 3.5%, CRISP 2.0%. Final composition (63 entries): Kunitz 21, PLA2 16 (including P86368 recovered at the VenomZone cross-check), snaclec/CTL 8, SVMP/disintegrin 5, SVSP 4, CRISP 4, LAAO 2, VEGF 2, NGF 1; species split *D. siamensis* 38 / *D. russelii* 25. Structure availability: 11 PDB entries, 48 AlphaFold models, 4 without structure.

SM.2 Six-source disease-gene integration (Phase 1B)

All retrievals 2026-09-22 (OMIM 2026-09-24). Phenotype scope: thrombotic microangiopathy (TMA), thrombotic thrombocytopenic purpura (TTP), atypical/typical hemolytic uremic syndrome (aHUS/HUS), microangiopathic hemolytic anemia (MAHA), snakebite envenoming, and venom-induced consumption coagulopathy (VICC). snakebite envenoming/VICC have no entries in Open Targets, DisGeNET, or OMIM (verified); the snakebite-specific layer is therefore carried by the toxin-target edge table (SM.3).

- Open Targets Platform (GraphQL API): 8 MONDO terms (TMA; TTP ×3; aHUS ×3; HUS), association score $\geq 0.05 \rightarrow 2,014$ records $\rightarrow 741$ unique genes (backbone).
- DISEASES (JensenLab knowledge file, curated channels UniProtKB-KW/MedlinePlus): 15 core genes, all overlapping.

- GeneCards (manual browser export, six terms; per-term median score thresholds — TMA 1.39, aHUS 113.85, TTP 47.79, MAHA 107.24, snakebite 20.40, VICC 46.30): 1,356 unique genes, 698 above threshold, 1,075 newly merged.
- DisGeNET (academic API, CURATED sources only; 16 UMLS CUIs incl. C2717961/C0034155/C2931788/C0019061; score ≥ 0.1): 46 unique genes, all already present; 2 entries (score ≥ 0.3) promoted to the core set.
- CTD (batch query, manual export; disease whitelist: Snake Bites / aHUS / HUS / TTP incl. acquired / Thrombotic Microangiopathies / Anemia, Hemolytic to block CTD term expansion): direct-evidence genes $\rightarrow$ core (80 unique, 3 new); InferenceScore $\geq 50 \rightarrow$ broad set (349 new); snakebite-envenoming term fully retained (57 genes, 13 new).
- OMIM (API; 44 MIM entries from five phenotype terms, phenotype whitelist of 14 entries; mappingKey = 3 only): 15 genes, all already in the core set.

Nine a priori four-axis molecules not covered by any source were force-included per the documented analysis plan: F2, F5, F10, FGG, PROC, VCAM1, ICAM1, SELE, HAVCR1 (sensitivity analysis in Table S6). Core-set rule: GeneCards $\geq$ per-term median $\cup$ OpenTargets score $\geq 0.3 \cup$ present in ≥ 3 disease terms $\cup$ DISEASES curated $\cup$ DisGeNET $\geq 0.3 \cup$ CTD direct $\cup$ OMIM (mappingKey 3) $\cup$ a priori molecules. Final library: 2,190 genes (core 1,042 / broad 1,148); all 27/27 a priori molecules are in the core set. Symbols were mapped to human UniProt AC with mygene (species=human).

SM.3 Toxin–host edge mapping and four-tier evidence (Phase 2)

Edges were built in four evidence tiers. L1 (curated interaction databases CTD/T3DB/BindingDB): 0 edges — CTD blocked programmatic access (ALTCHA) and no equivalent curated toxin–target rows exist for these toxins. L2 (literature-curated): 49 edges / 15 unique human targets, extracted from UniProt/Swiss-Prot curated FUNCTION annotations of the 63 entries (43 entries with target-related descriptions) plus targeted PubMed verification (PMIDs 27089306, 21356226, 18554518, 28042812, 36423674, 37092784, 28732041, 21871889, 18222185, 21802487); family-level edges are flagged (score 0.6) when a publication could not be assigned to a specific library entry. L3 (BLAST homology auxiliary): suspended (no local blastp; plan-labelled auxiliary only). L4 (STRING one-hop expansion): for the 15 direct targets, STRING API interaction_partners (H. sapiens, combined score ≥ 0.7 , top 10) gave 105 unique neighbours $\rightarrow$ 398 edges. Total 447 edges / 112 unique targets. Entries without curated human targets are recorded as no-edge with reasons (CRISP $\times 4$, LAAO $\times 2$, chaperone PLA2 P31100, snaclec Q4PRC6–9/Q4PRD0).

SM.4 Network construction, topology, and robustness (Phase 3)

Intersection: 15 direct targets (L1+L2) $\cap$ 1,042 core disease genes $\rightarrow$ strict core of 10 (COL4A1, F10, F11, F5, FGA, GP1BA, KDR, PLG, PROC, PROS1). An extended sensitivity/topology layer was built as (strict core $\cup$ L4 STRING neighbours $\cap$ core set) = 54 nodes. PPI edges were exported from the STRING API (string-db.org/api, STRING v12.0, queried 2026-09-24) at combined score ≥ 0.7 (main; 321 edges) and ≥ 0.4 (sensitivity; 615 edges). Hub ranking used a Python re-implementation of cytoHubba Degree, MCC (maximal clique centrality) and EPC (edge percolated component; $p = 0.5$, 1,000 simulations, seed 42), Top-20 consensus $\rightarrow$ 10 hub genes. EPC had low discrimination on this dense coagulation network (scores converge; stated in Methods), so the consensus is driven by MCC/Degree. Robustness: targeted removal in static degree order (computed once on the full

network) versus 100 random-removal shuffles (seed 1234); curves include $k = 0$ (LCC/N = 1.0) as the first point; at 27/54 nodes removed (50%), LCC = 0.241 (targeted) vs 0.480 ± 0.023 (random).

SM.5 Dual-platform enrichment (Phase 4)

Input: Hub 10 $\cup$ strict core 10 = 15 unique genes. Platform 1: g:Profiler g:GOST (API, 2026-09-24; organism *hsapiens*; sources GO:BP/CC/MF + KEGG + Reactome; g:SCS correction, user threshold 0.05; no custom background) $\rightarrow$ 116 significant terms. Platform 2: Enrichr (KEGG_2021_Human, Reactome_2022, GO_Biological_Process_2023; BH correction) $\rightarrow$ 202 significant terms. Cross-platform correspondences are theme-level (not identical term IDs); nine themes matched by keyword. A re-run of the 15-gene g:Profiler query on 2026-10-02 reproduced the archived 116-term set exactly (reproducibility check, Table S6).

SM.6 Structure validation and small-molecule docking (Phase 5)

Structures (Table S7): experimental PDB entries 2E3X (RVV-X heavy chain), 3S9C (RVV-V γ -FV complex), 1KPM (PLA2), 1WQ9 (svVEGF), 2W26 (FXa), 7KVE (FV), 3GHG (fibrinogen fragment), 3V2A (VEGFR2), 1QRZ (plasmin), 1PPB; AlphaFold models (pLDDT > 70 rule) for daborhagin-K, Kunitz inhibitor, snaclec, and 1M10 was used only before retraction (see below). ClusPro (Balanced mode, blind docking): pre-specified acceptance = top-1 cluster $\geq$ 30 members; 7/7 pairs passed (59–190). HADDOCK 2.4 (biochemical AIR restraints; interface cutoff 5 Å): pre-specified double pass = score $\leq$ -100 and functional-site coverage; P2 passed both (score -144.9 ± 4.4 , $Z = -1.4$, 6/6); P1/P3/P6 achieved restraint orientation (nearest restraint distance 1.8–7.8 Å) without reaching the score threshold and are reported as site-orientation evidence only. HDock (template docking, P7): confidence 0.8969 (> 0.7), 21/26 reference-interface coverage, receptor superposition on 3V2A RMSD 0.000 Å. Orthogonal descriptors (post hoc, labelled): iMODS NMA mobility ratios and PRODIGY $\Delta G/K_d$ for the 7 pairs.

Retraction note: pair P4 (snaclec $\times$ "GP1BA") was withdrawn on 2026-10-02 after the receptor chain was found misidentified — 1M10 chain A is the VWF-A1 domain, not GP1BA; the pair is removed from the evidence chain (manuscript protocol-deviation table; Fig. 6 panel D blanked).

Small molecules (Table S8): AutoDock Vina 1.2.7 + Meeko 0.8.0; ligands from PubChem 3D (batimastat CID 5362422, marimastat CID 119031, varespladib CID 155815); receptors RVVX_A (2E3X chain A, catalytic Zn retained, +2), daborhaginK (AlphaFold + modelled Zn at His339/343/349 — arm not trusted), PLA2_1KPM (chain A, grid centred on His48/Asp49); grid box $26 \times 26 \times 26$ Å (centres in Table S8), exhaustiveness = 8, num_modes = 9, default seed. Hydroxamate Zn-chelation geometry was verified on the top poses (Table S8).

SM.7 GEO transcriptome re-analysis (v41 canonical pipeline)

GSE121297 (human HUVEC $\times$ rhodocetin- $\alpha\beta$; Affymetrix HG-U133 Plus 2.0 / GPL570): static RCab (GSM3430997–99) vs static control (GSM3430994–96), 3 vs 3; the 80-rpm arm was not used. GSE248215 (mouse skeletal muscle, *D. russelii* venom 30 μ g in vivo; NanoString Fibrosis V2, 760 genes): DR 24 h (3) vs PBS control (2); author-normalised log2 matrix used. Differential expression: Welch's t-tests in Python with Benjamini–Hochberg correction (script `geo_v41_reanalysis.py`). Probe handling: one fixed probe per gene, highest mean across the six contrast samples chosen before testing; a probe mapping to several symbols contributes to each symbol $\rightarrow$ 22,453 genes. An earlier

archived run (21,183 genes; multi-symbol probe conflicts resolved differently) gives identical log2FC/raw P for all shared genes and the same conclusion: 0 genes at BH-adjusted $P < 0.05$ in either dataset. All interpretation is nominal, exploratory, and labelled as such; the two tissues are not combined into a single validation cohort.

SM.8 Protocol deviations (mirrors manuscript Table 1)

Item	Original documented plan	Final analysis	Reason for change	Results seen before change?
Hypotheses	H1 SVMP–VWF/ADAMTS13 dominant; H2 complement & endothelial enrichment; exploratory H3	H1 coagulation/platelet prominence; H2 complement as secondary host response; H3 dropped	Intersection and enrichment results motivated reformulation	Yes — reformulated hypotheses are post hoc and labelled
Module screening	MCODE cluster screening	MCC/Degree/EPC Top-20 consensus (re-implementation)	MCODE step not executed in the final pipeline	Yes
Docking controls	3–5 negative docking pairs	Not performed	Resource constraints; listed as a limitation	n/a
Enrichment platforms	Metascape + clusterProfiler	g:Profiler + Enrichr	Web-platform accessibility and full-request auditability	Yes
GEO statistics	limma (R)	Welch's t-tests + BH in Python	Python-only analysis environment; method renamed accordingly	Yes
Orthogonal corroboration	Not planned	PRODIGY + iMODS model-property descriptors	Added during revision; reported as post hoc descriptors	Yes (post hoc addition)

SM.9 Reproducibility manifest

- Environment: Python 3.12 (pandas, numpy, scipy, matplotlib, seaborn, networkx, openpyxl); web services — ClusPro, HADDOCK 2.4, HDock, iMODS, PRODIGY, g:Profiler, Enrichr, STRING v12.0; AutoDock Vina 1.2.7 + Meeko 0.8.0; PyMOL (figures); Cytoscape 3.x (Fig. 5 rendering).
- Fixed random seeds: 1234 (robustness shuffles), 42 (EPC percolation); all pipeline scripts under scripts/ and phase directories; reproduction order (14 steps) in the repository README.
- Archive: GitHub <https://github.com/Wenhui-cangku/snakebite-TMA-network-toxicology> · Zenodo concept DOI 10.5281/zenodo.23065792 (resolves to the latest release) · large raw files dataset 10.5281/zenodo.23087108.
- Change log: version-by-version record v1–v47 (CHANGELOG.csv in the repository; public copy of the data-management table in 00_protocol/).
- Reproducibility checks passed: 15-gene g:Profiler rerun = archived 116 terms (2026-10-02); robustness script rerun reproduces archived curves value-by-value; GEO script reproduces the canonical 22,453-gene table.

Supplementary Tables S1–S10

Full machine-readable tables are provided in the companion workbook Supplementary_Tables_S1-S10.xlsx (one sheet per table; sheet names in parentheses). Key tables are also reproduced inline below when compact.

Table S1. Venom-toxin master table (sheet S1_Toxin_master; 63 entries). entry_id, toxin_name, toxin_family, UniProt/GenBank ID, entry_name, species, sequence length, source (incl. unreviewed/NCBI flags), cluster ID (40 clusters, 90% identity), structure status (PDB/AlphaFold/none), family-level venom abundance weight (%), inclusion status and exclusion reason, download date. Companion FASTA files (all_toxins_raw.fasta, toxins_nr90.fasta) are in the Zenodo raw dataset. Cited in Methods §2.1 and Fig. 1.

Table S2. Integrated disease-gene library (sheet S2_Disease_genes; 2,190 records). gene_symbol, UniProt AC, Entrez ID, per-source evidence (GeneCards score, DisGeNET score, OMIM yes/no, CTD inference, OpenTargets score, OT disease count, DISEASES curated), core-set flag (1,042 core / 1,148 broad), pathology-axis assignment (1–4), source query terms, download date. Cited in Methods §2.2 and Fig. 2.

Table S3. Toxin–host edge table with four-tier evidence (sheet S3_Toxin_target_edges; 447 edges). toxin_id, toxin_ac, family, target_symbol, evidence_type (L2 literature-curated / L4 STRING neighbour), evidence_detail (effect direction, isoform/complex notes, experimental context, or “via <target>” for L4), PMIDs, evidence score, download date. L1 = 0 and L3 suspended (SM.3). Cited in Methods §2.3 and Figs. 2–5.

Table S4. Full three-algorithm hub ranking (sheet S4_Hub_full_ranking; 54 nodes). Degree, MCC, and EPC scores with competition ranks (ties share the minimum rank, reported in full) for every node of the extended layer, plus hub-10 / strict-core-10 flags and pathology-axis labels. Network settings in SM.4; EPC low-discrimination note applies. Cited in Results (hub analysis) and Table 2.

Hub-10 excerpt (full 54-node ranking in the workbook):

gene_symbol	Degree	MCC	EPC	pathology_axis	strict_core
F2	25	1506956.0	52.22	Axis2	no (extension layer)
F3	22	1502082.0	52.22	—	no (extension layer)
ALB	17	12158.0	52.22	—	no (extension layer)
F8	16	1451808.0	52.22	—	no (extension layer)
F7	14	1493280.0	52.22	—	no (extension layer)
PLG	21	20030.0	52.22	Axis2	yes
PROC	18	1130700.0	52.22	Axis2	yes
FGA	16	13230.0	52.22	Axis2	yes
F10	15	1497144.0	52.22	Axis2	yes
F5	14	725838.0	52.22	Axis2	yes

Table S5. Enrichment results (sheets S5a_gProfiler, S5b_Enrichr, S5c_Prior_pathways). S5a: all 116 g:Profiler terms significant at g:SCS < 0.05 (15-gene input) with term IDs, sizes and driver genes. S5b:

all 424 Enrichr rows across the three libraries, including non-significant terms (rank, P, BH q, overlap, genes). S5c: the seven a priori pathway tests on both platforms with conclusions and driver genes. Cited in Results §3.4 and Fig. 4 / Table 3.

prior_pathway	g:Profiler (platform 1)	Enrichr (platform 2)	conclusion	notes/driver_genes
hsa04610 Complement and coagulation cascades	KEGG p=3.90e-18 ✓	adj=1.63e-22 ✓	significant on both platforms ✓✓	all 11 driver genes are on the coagulation/fibrinolysis arm (F2/F3/F5/F7/F8/F10/FGA/PLG/PROC/F11/PROS1); zero genes on the complement arm — key point for H2
hsa04611 Platelet activation	REAC equivalent term p=1.47e-07 ✓	adj=1.45e-03 ✓	significant on both platforms ✓✓	GP1BA/ALB/F2/F5/F8/FGA/PLG/PROS1
ECM–receptor interaction	did not reach g:SCS<0.05	adj=1.81e-02 ✓	single platform (Enrichr)	driven by COL4A1 et al.
Focal adhesion	did not reach g:SCS<0.05	adj=4.68e-02 ✓	single platform (Enrichr, borderline)	—
PI3K–Akt	not significant	adj=9.08e-02	not significant	—
NF-κB	not significant	no such term	not significant	Axis-4 inflammatory dimension has no support on the 15-gene core set; recorded as-is
Fluid shear stress & atherosclerosis	not significant	adj=1.52e-01	not significant	—

Table S6. Sensitivity analysis without forced inclusion (sheets S6_Sensitivity, S6_gProfiler_11genes).

Documents the effect of removing the nine force-included a priori genes: input sets, disease-set sizes (1,042 → 1,033 core), strict-core/hub composition, and a full g:Profiler rerun on the 11-gene input (93 terms). The dominant coagulation/platelet theme is qualitatively stable (fibrin clot 7/11, P = 3.8e-13; KEGG:04610 7/11, P = 1.4e-10); gamma-carboxylation becomes marginal. The sheet also records the 15-gene reproducibility rerun (116/116 terms identical). Cited in Results §3.2.

Table S7. Structure-validation summary (sheets S7_Docking, S7_iMODS_PRODIGY). Seven toxin–host pairs across ClusPro (blind, Balanced), HADDOCK 2.4 (AIR-restrained) and HDock (template): cluster sizes, energies/scores, Z-scores, interface residue counts, functional-site coverage, restraint distances, and verdicts, plus the iMODS/PRODIGY orthogonal descriptors. The retracted P4 pair is listed as a marked retraction row. AIR restraint files, PDB/mmCIF inputs and output models are in the Zenodo raw dataset. Cited in Results §3.5–3.6, Figs. 6–7, Tables 4–5.

Compact view of the iMODS/PRODIGY orthogonal descriptors (full ClusPro/HADDOCK/HDock metrics and notes in the workbook):

pair	complex	PRODIGY ΔG (kcal/mol)	Kd (M)	iMODS λ1	interface_residues(5Å)	interface/rest_mobility_ratio
P1	RVV-X×FXa	-13.8	8e-11	2.864e-05	71	0.53
P2	RVV-Vy×FV	-11.7	2.8e-09	2.671e-05	64	0.88
P3	dabK×FIB	-15.2	7e-12	2.605e-07	46	0.78
P4	snaclec×GP1BA	-9.2	1.9e-07	1.004e-06	34	0.52
P5	PLA2×FXa	-13.7	8.5e-11	0.0002564	63	0.86
P6	Kunitz×plasmin	-10.7	1.5e-08	1.485e-05	55	0.81
P7	svVEGF×VEGFR2	-12.1	1.3e-09	6.553e-05	48	0.52

Table S8. Small-molecule docking (sheet S8_Small_molecule). Nine receptor × ligand results (batimastat/marimastat/varespladib × RVV-X/daborhagin-K/PLA2): best Vina affinities, count of modes ≤ -7.5 kcal/mol, grid centres (26 Å box), and the Zn-chelation geometry check; the daborhagin-K AlphaFold + modelled-Zn arm is marked not trusted. Settings in SM.6. Cited in Results §3.7 and Fig. 8.

Table S9. GEO re-analysis (sheets S9a_GEO_samples, S9b_GSE121297_DEG, S9c_GSE248215_DEG, S9d_GEO_watchlist). S9a: sample tables and contrast design (GSM IDs, titles, platforms, groups used). S9b/S9c: full per-gene Welch + BH tables (22,453 and 760 genes; v41 canonical fixed-probe pipeline). S9d: the 35-molecule a priori watchlist across both datasets (log2FC, P, significance flags; ALB and COL4A1 not measured by either platform). Cited in Results §3.8, Figs. 9–10, Table 6.

Table S10. Network robustness (sheets S10_Robustness, S10_Removal_order). $k = 0 \dots 54$ removal axis ($k = 0$ included); the static-degree targeted curve, all 100 per-run random-removal curves (seed 1234), and mean $\pm$ SD columns; the targeted removal order with degrees. Regeneration script: 04_network/robustness_analysis_v41.py. Cited in Results §3.9 and Fig. 11.

Companion data files

- Zenodo raw dataset (10.5281/zenodo.23087108): toxin FASTA files; PDB/mmCIF docking inputs and output models (ClusPro/HADDOCK/HDock); GEO raw matrices and GPL570 probe annotation; iMODS/PRODIGY raw outputs; CTD raw TSV exports; file inventory in README_DATA.md inside the zip.
- Code + result tables: GitHub repository and Zenodo software archive (concept DOI 10.5281/zenodo.23065792, resolving to the latest release), including the change log (v1–v47) and the public data-management table.
- Cytoscape session files for Fig. 5 (cyjs/xgmml) and the vector SVG export: included in this package as Figure5_4layer_network.svg; source session files in repository 04_network/cytoscape/.
- Operational SOP for future MD/free-energy work (planned, not executed): repository 08_md/MD_SOP.md — explicitly a future-work protocol, not completed evidence.